

5	C	Applying Newton's second law on the system of both crates, $F_{net,both} = m_{both}a$ $100 - (2.0 + 3.0)(9.81) = (2.0 + 3.0) \times a$ $a = \frac{100 - (2.0 + 3.0)(9.81)}{(2.0 + 3.0)} = 10.19 \text{ m s}^{-2}$ Applying Newton's second law on the 2.0 kg crate, $F_{net,2kg} = m_{2kg}a$ $100 - 2.0(9.81) - T = 2.0(10.19)$ $T = 60 \text{ N}$ OR $F_{net,3kg} = m_{3kg}a$ $T - 3.0(9.81) = 3.0(10.19)$ $T = 60 \text{ N}$
6	B	Motorcycle travels in the same direction during the whole duration